Integrated Aero-Structural Optimization of Bio-Inspired and Generative AI Airfoils with Internal Lattice Architectures

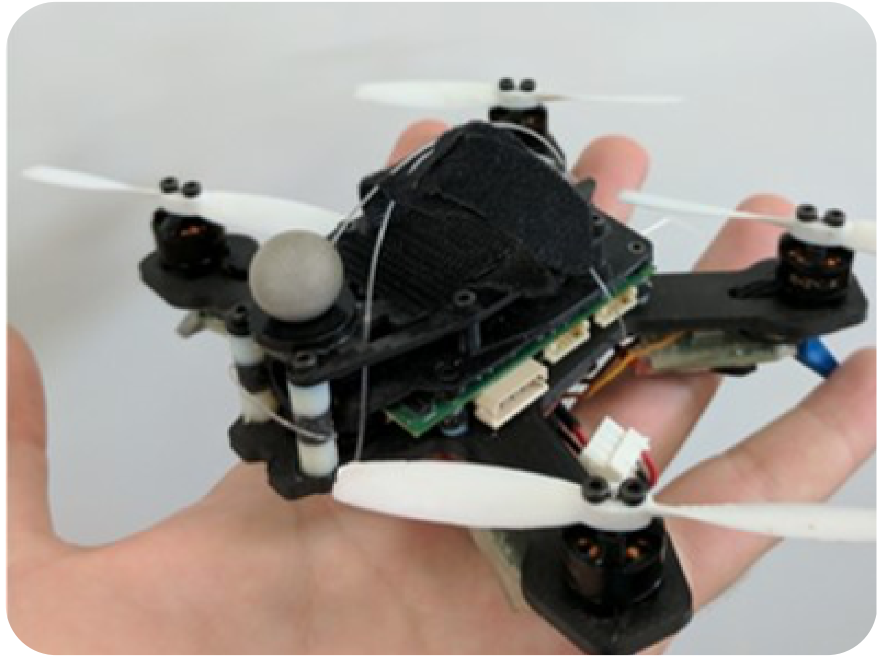
AIAA Region II Conference Presentation

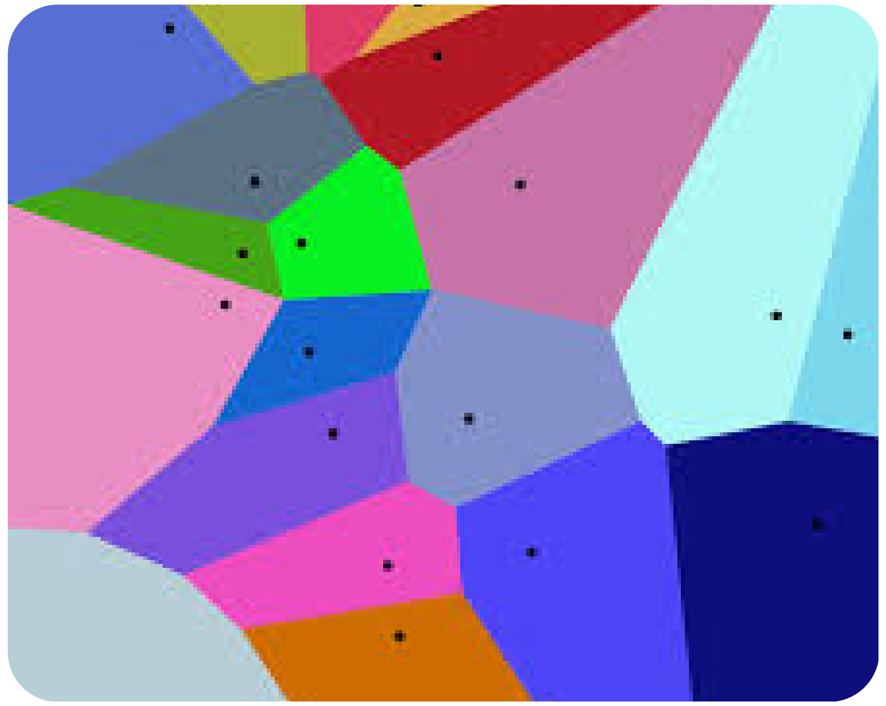
Principal Findings and Presentation Roadmap

- Low-AR finite wings diverged from 2D predictions
- GOE 265 gave best measured performance
- Stability-limited GenAI and NACA 5415
- Bio-inspired lattice improved structural survivability
- Talk flow: motivation, method, results, implications

Research Motivation and Design Context

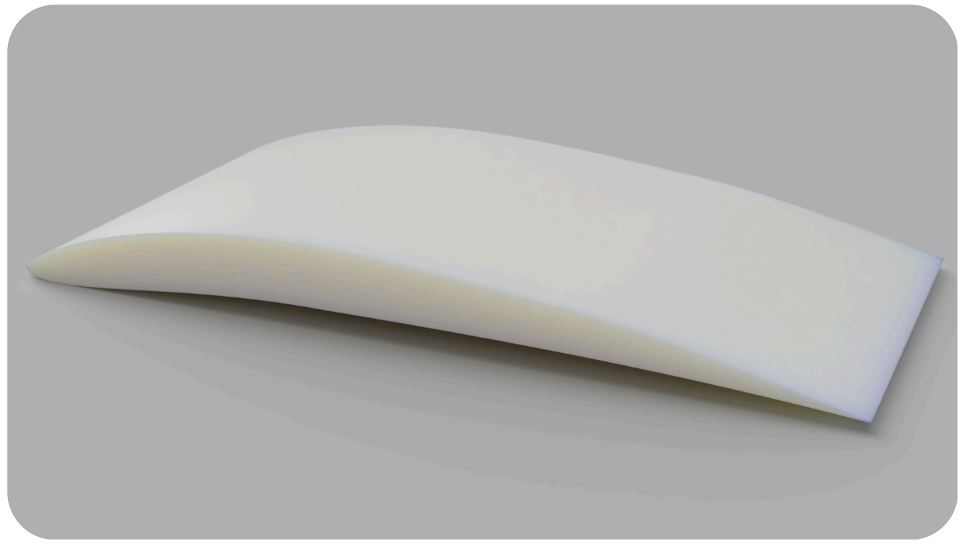
- MAV and sUAS wings operate at low Reynolds number
- Short printed wings amplify 3D effects
- Induced drag can dominate section behavior
- Mounting compliance alters measured response
- Internal structure influences twist, stiffness, stability

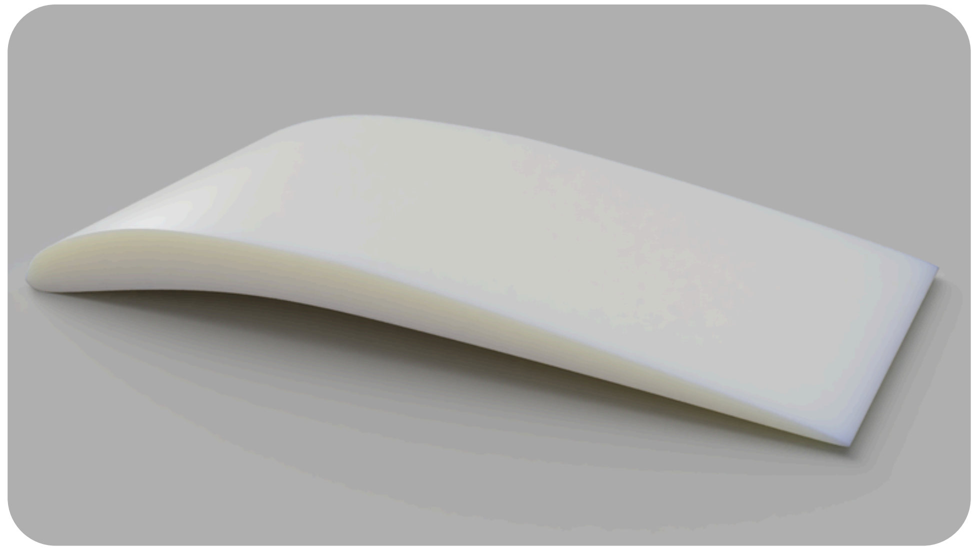


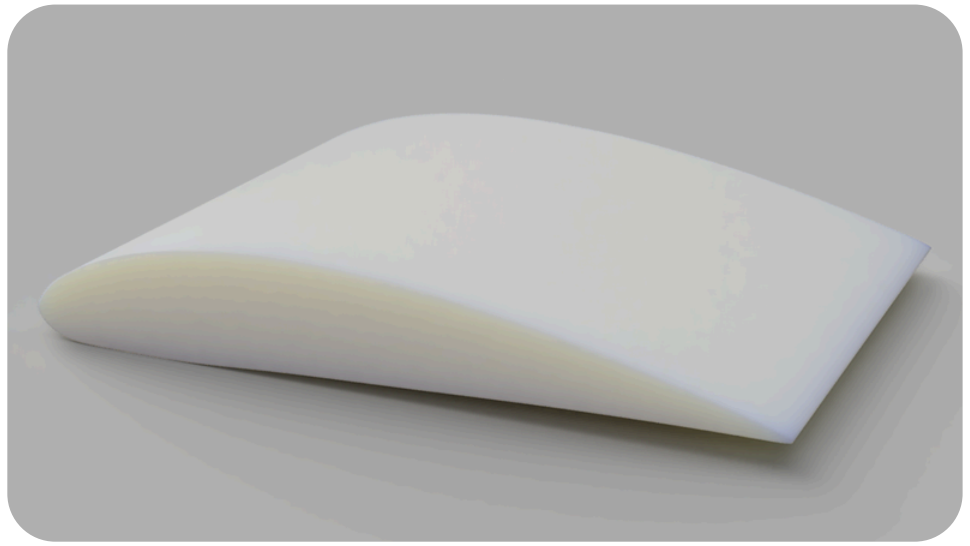


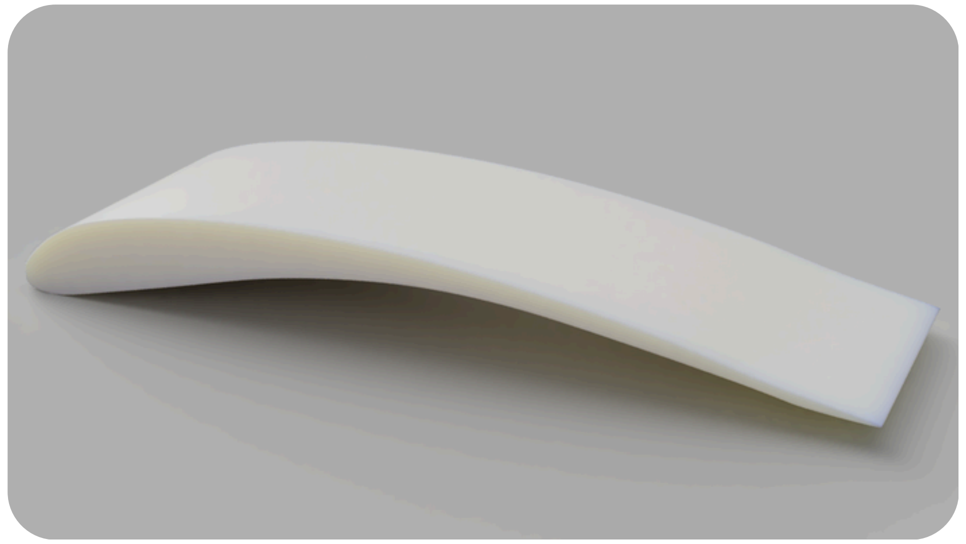
Objectives, Test Articles, and Evaluation Scope

- Compare finite-wing performance at $\alpha = 0°$
- Assess force trends, L/D, and stability limits
- Benchmark against NeuralFoil section predictions
- Tested: GenAI, GOE 265, NACA 5415
- S1223 included computationally only


(a) GenAI airfoil wing render.


(b) GOE 265 airfoil wing render.


(c) NACA 5415 baseline wing render.


(d) S1223 wing render.

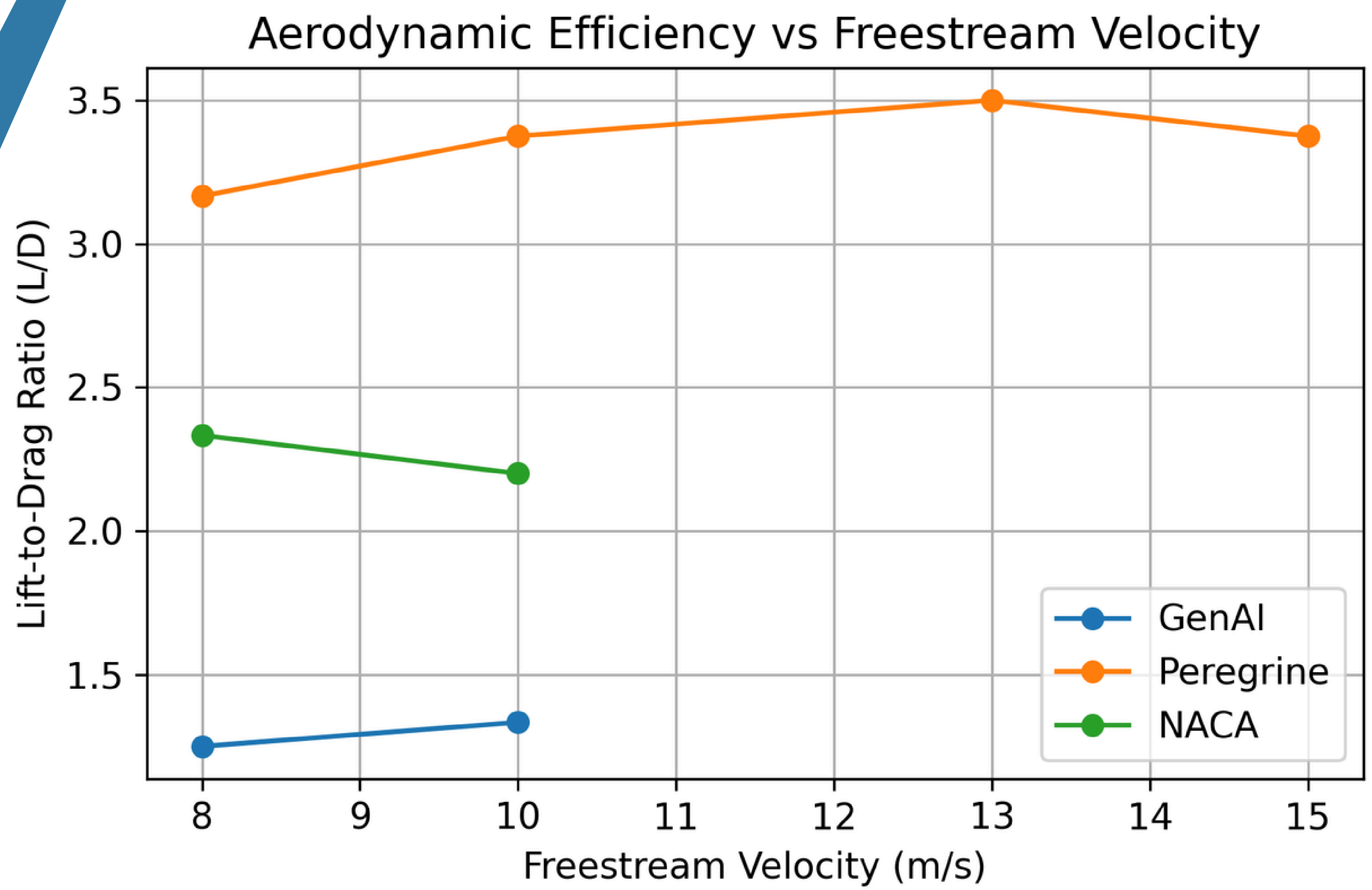
Span fixed at 3 in
Aspect ratios: 0.50, 0.50, 0.75, 0.32

Experimental Configuration and Stability Classification

- GUNT HM 170 open-circuit wind tunnel
- Two-component sensor measured lift and drag
- Test speeds: 8, 10, 13, 15 m/s
- Rigid mounting fixture held α = 0°
- “Unstable” = no clear steady reading

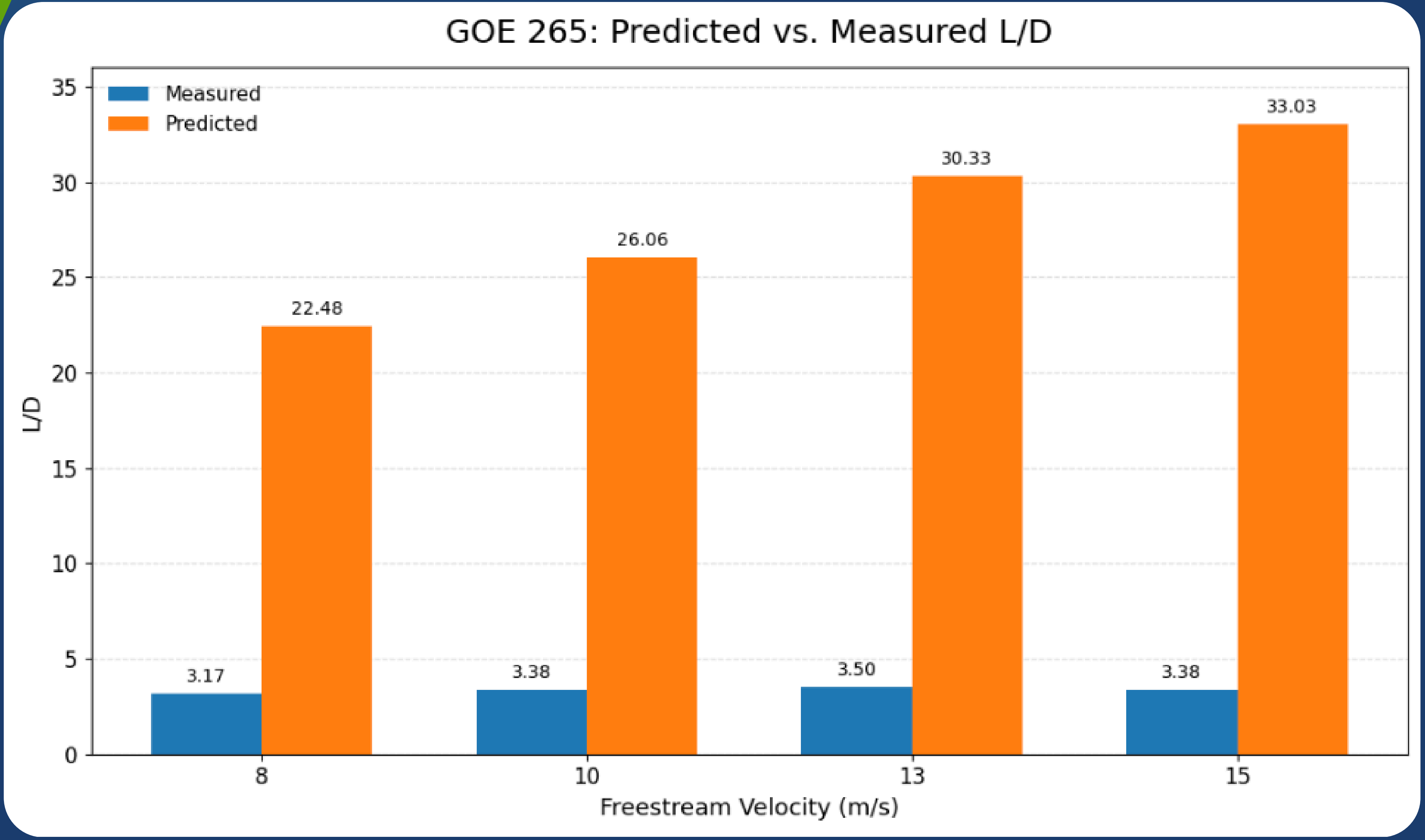
Measured Finite-Wing Aerodynamic Performanc

- GOE 265: stable from 8–15 m/s
- GOE 265 L/D: 3.17, 3.38, 3.50, 3.38
- GenAI: L/D = 1.25, 1.33
- GenAI unstable above 10 m/s
- NACA 5415 unstable above 10 m/s



Section Predictions Versus Finite-Wing Measurements

- NeuralFoil used as a 2D reference
- GOE 265 predicted L/D = 22.48–33.03
- GOE 265 measured L/D = 3.17–3.50
- Very low aspect ratio raised induced drag sharply
- Finite-wing validation proved essential



Stability as a Practical Design Discriminator

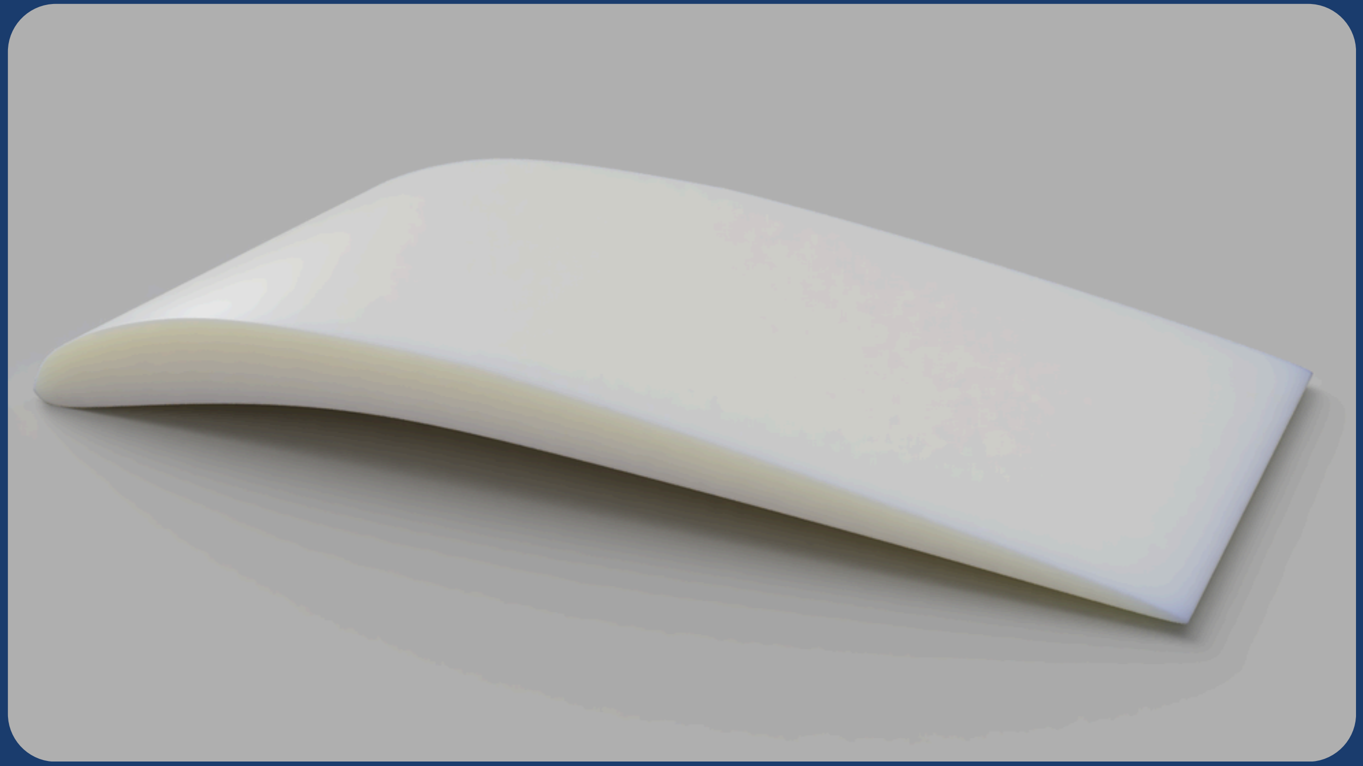
- GOE 265 was the best practical airfoil
- Stability separated the designs
- GenAI and NACA 5415 failed above 10 m/s
- Robustly stable can outperform theoretically superior

Bio-Inspired Internal Structural Architecture

Improving strength-to-weight performance without changing external aerodynamics

Motivation

- Findings in airfoil shape offer strong conclusions for optimized aerodynamic performance
- Optimizing the internal lattice/rib structure allows for strong resistance to stresses and strains
- Following the best-performing airfoil shape, the design methodology can follow a bio-inspired process
- Lattice structure can take inspiration from naturally occurring structures
- Machine learning-driven program can mimic “natural selection,” arriving at the strongest lattice configuration within the airfoil.

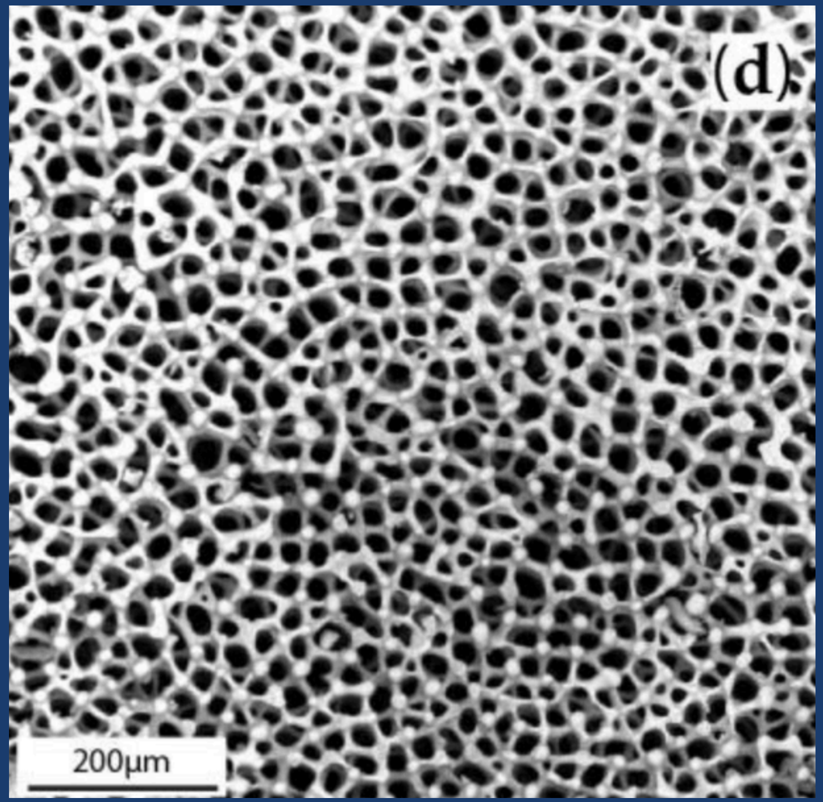


GOE 265 Airfoil

Lattice Design - *Bio-Inspiration*

- Structural Categories
  - Cellular
  - Fibrous
  - Sandwich

- Sea Urchin Stereom
  - Cellular/Sandwich
  - Voronoi Structure
  - Strength and Rigidity

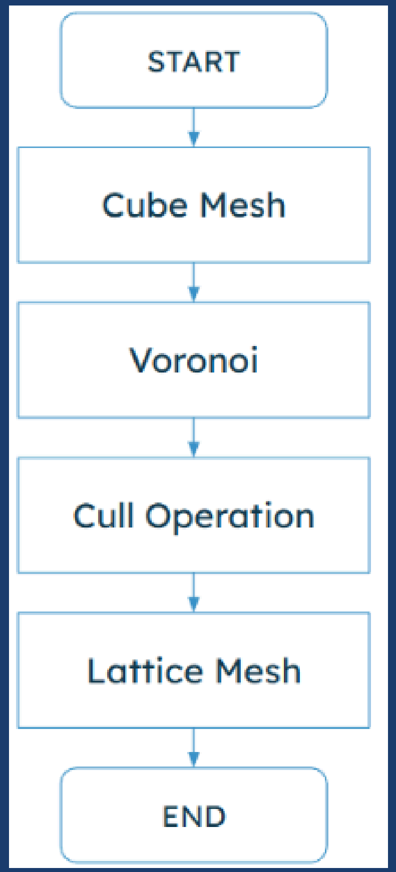


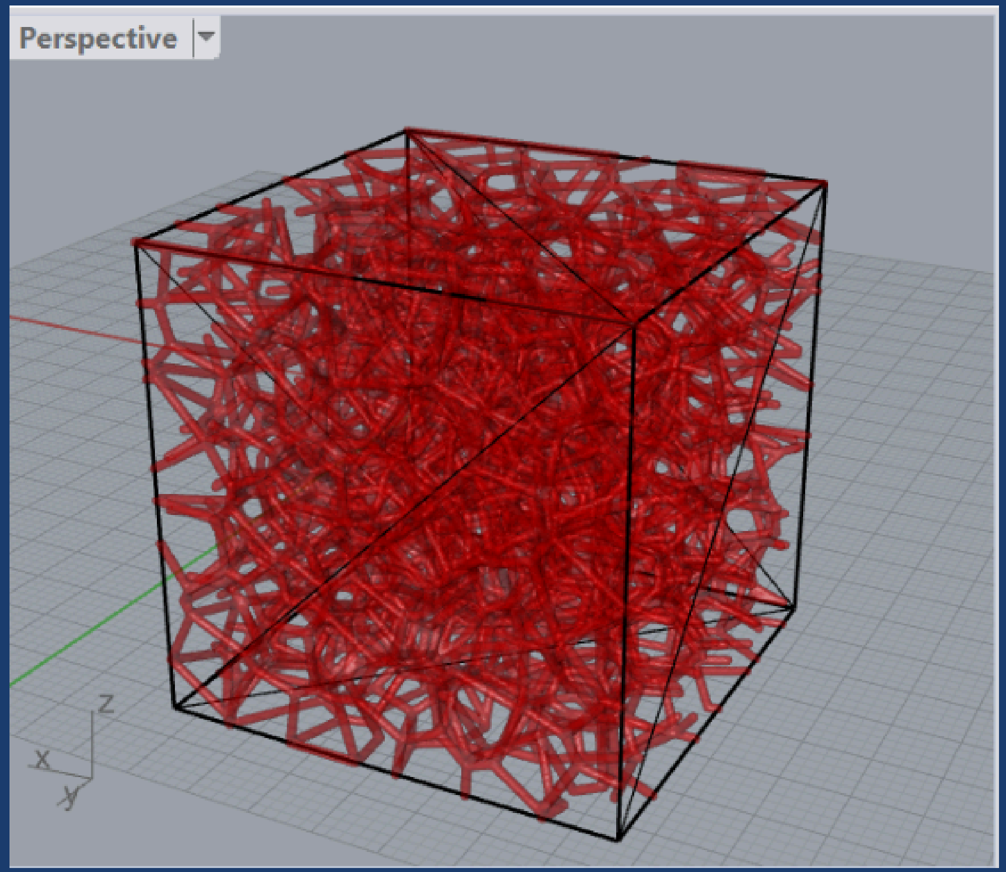
From:
Parametric Design and Mechanical Characterization of 3D-Printed PLA Composite Biomimetic Voronoi Lattices Inspired by the Stereom of Sea Urchins (Efstathiadis et al., 2022).

Lattice Design - *Voronoi*

- Rhinoceros 3D
  - Grasshopper
  - Voronoi Operation

- Voronoi Cube
  - Identify structurable/manufacturable patterns

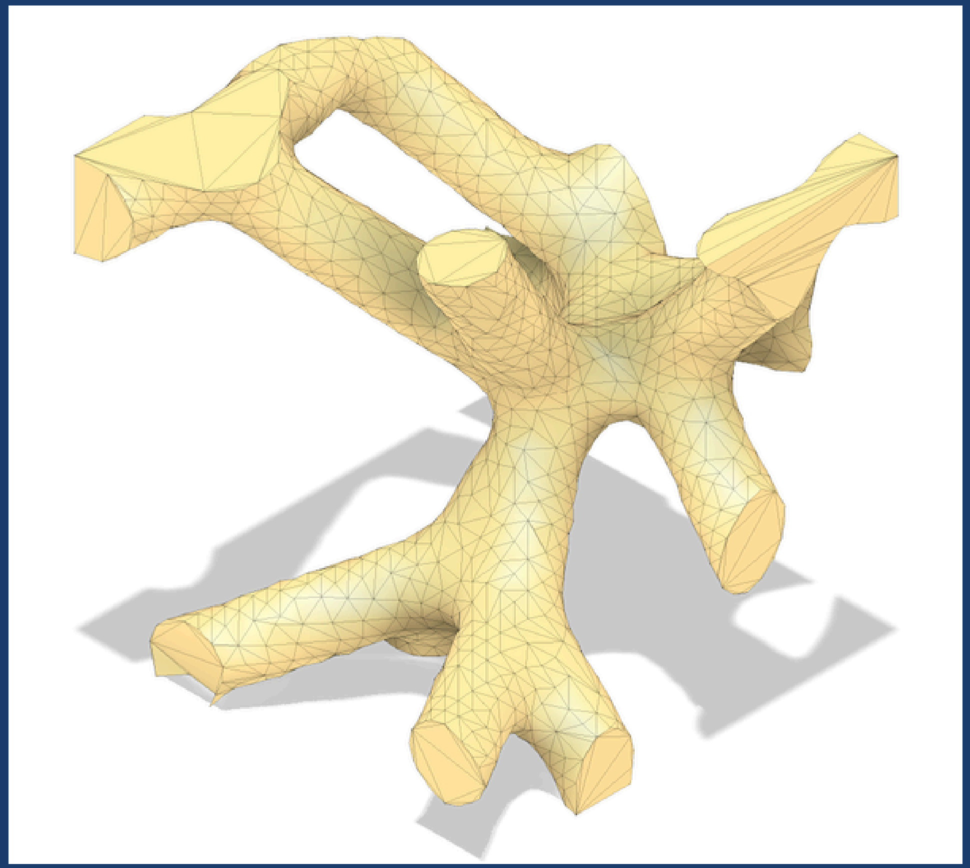




Lattice Design -
ROI Sampling

- MATLAB
  - Cubic Region of Interest (ROI)
  - Identify common patterns in samples

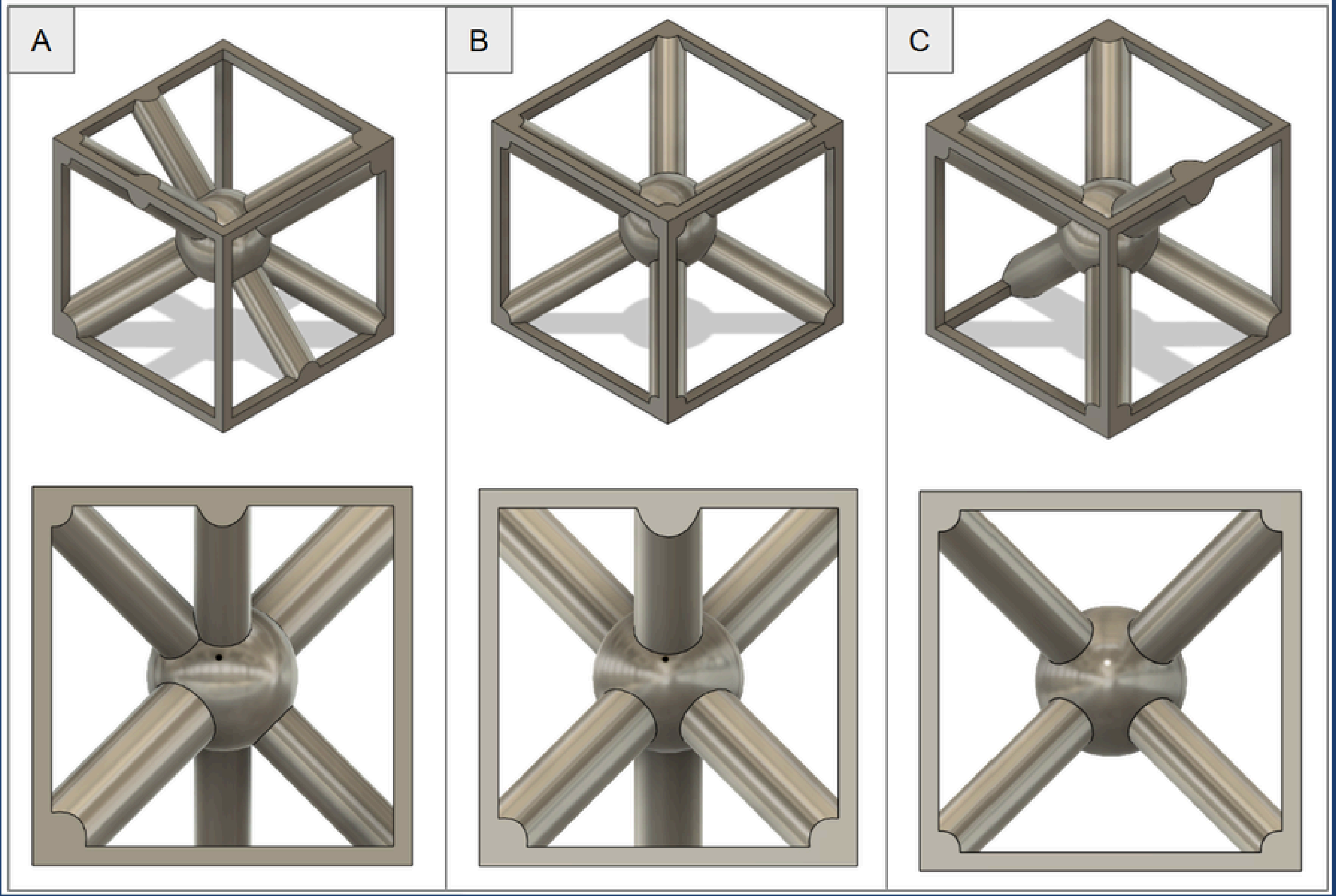
- Base Geometry
  - Core-centric
  - Side/corner load-bearing struts



Structure Modeling - *Unit Cube CAD*

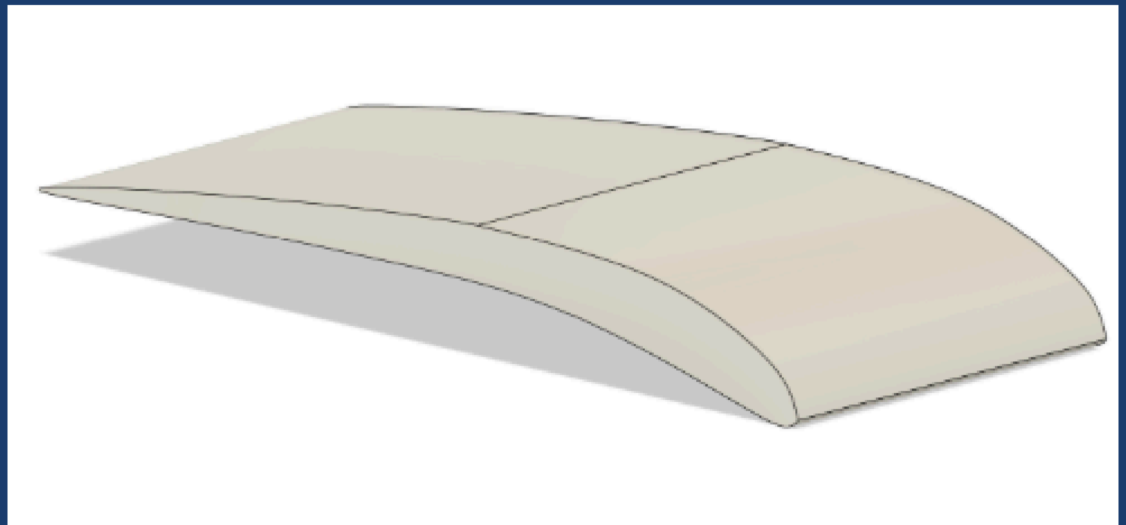
- Unit Cubes
  - Small, uniform samples
  - Variations supporting different loads

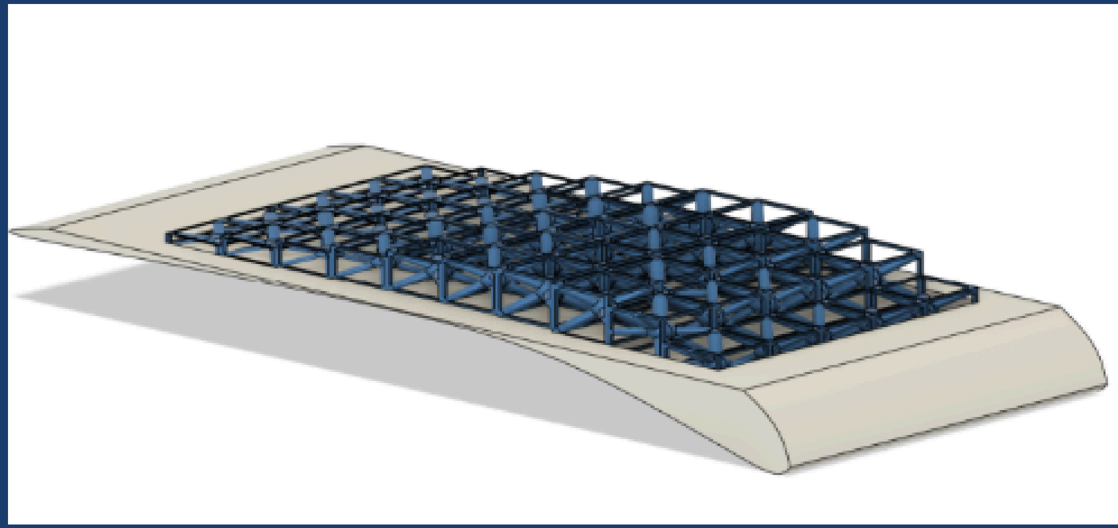
- The Marks
  - Core Centric
  - Mk II (A): Main diagonal strut
  - Mk III (B): Triple corner/side struts
  - Mk IV (C): Four corner struts

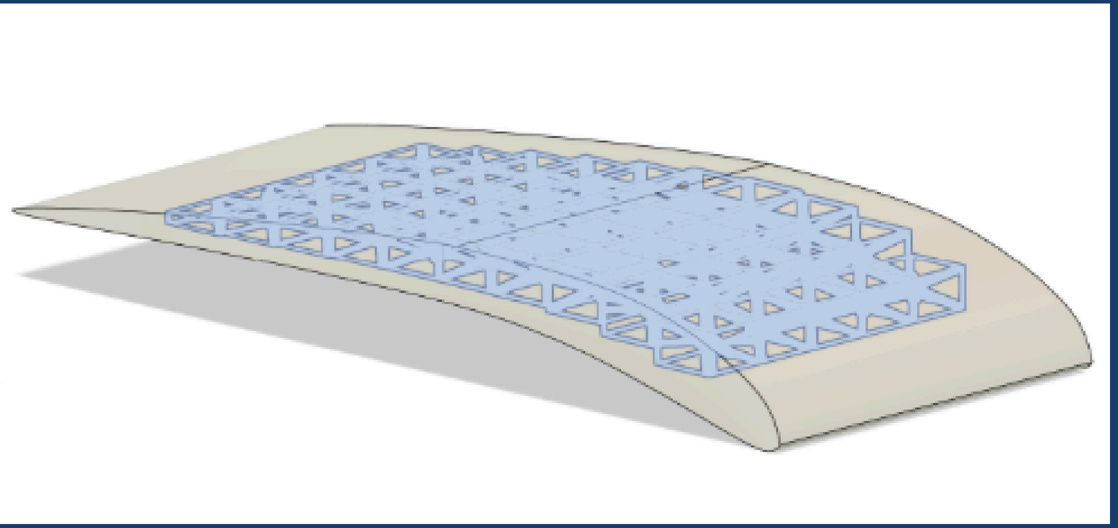


Structure Modeling -
Airfoil Lattice CAD

- GOE 265 Airfoil
- Unit cubes create lattice
- Cellular/sandwich structure

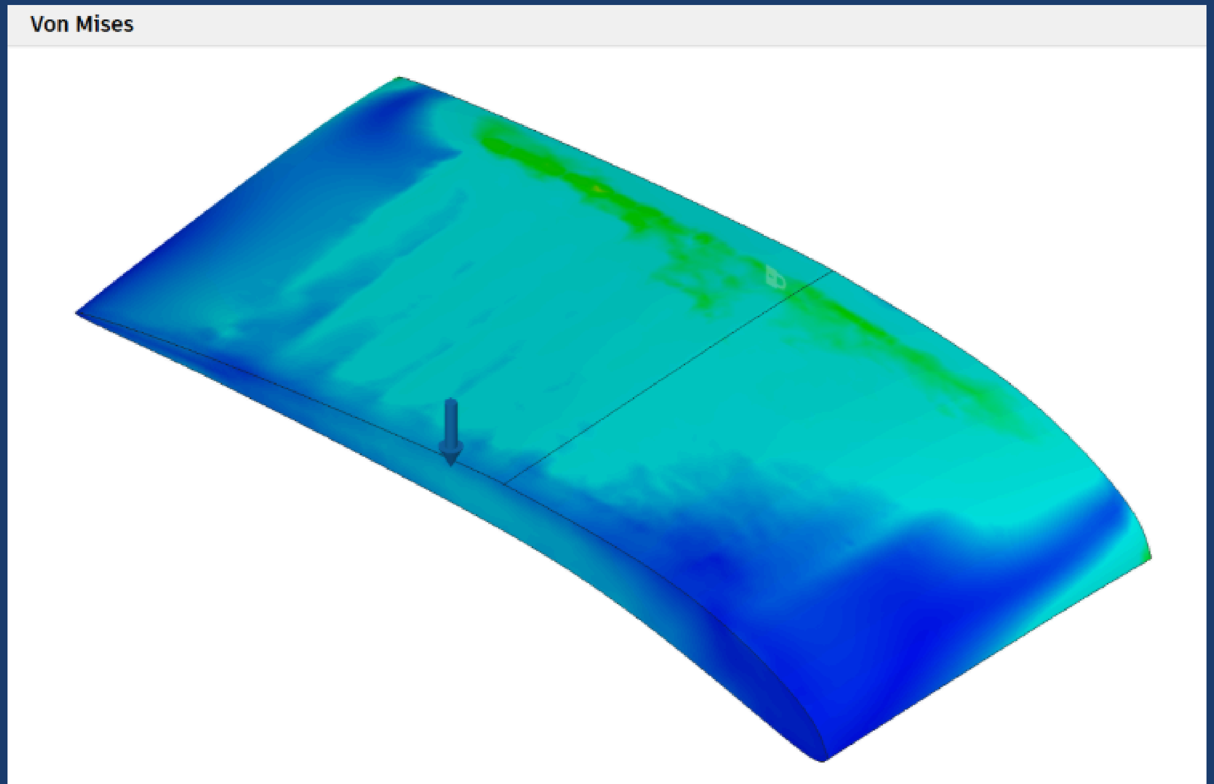


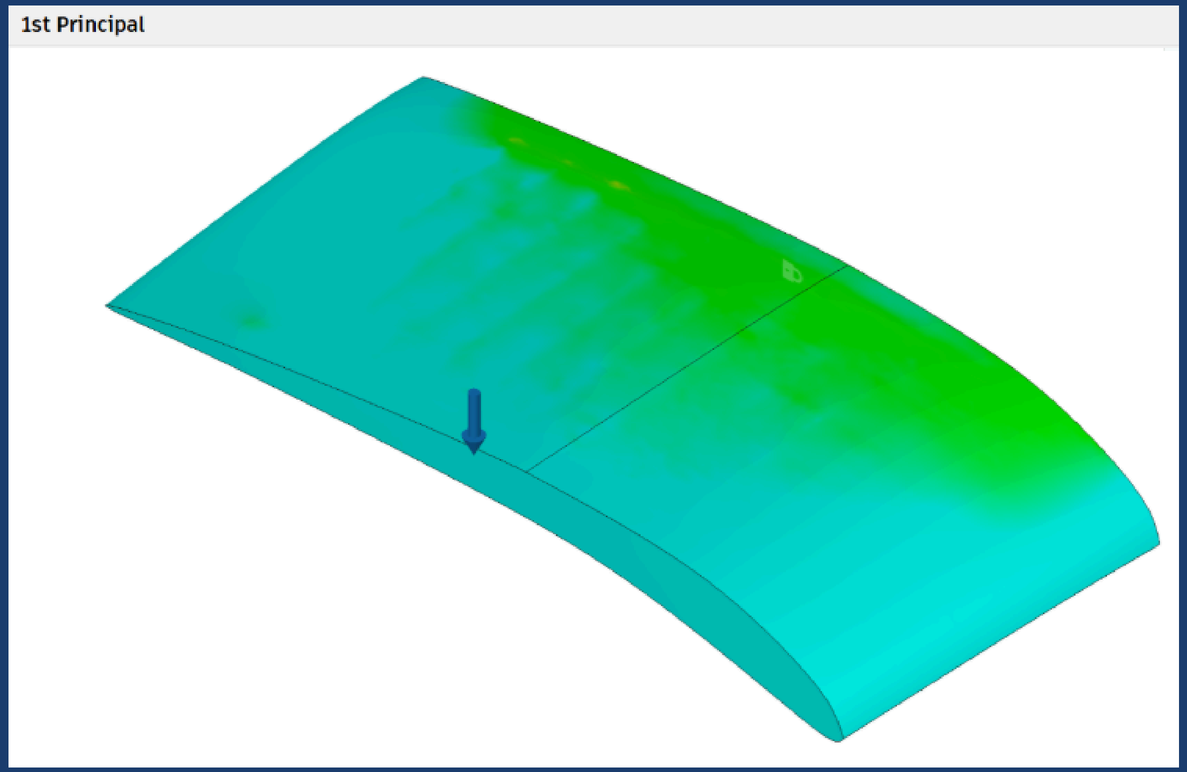




Simulation & Analysis - *Cantilever FEA*

- Lattice performance tested in Finite Element Analysis (FEA)
- Cantilever analysis demonstrates wing load

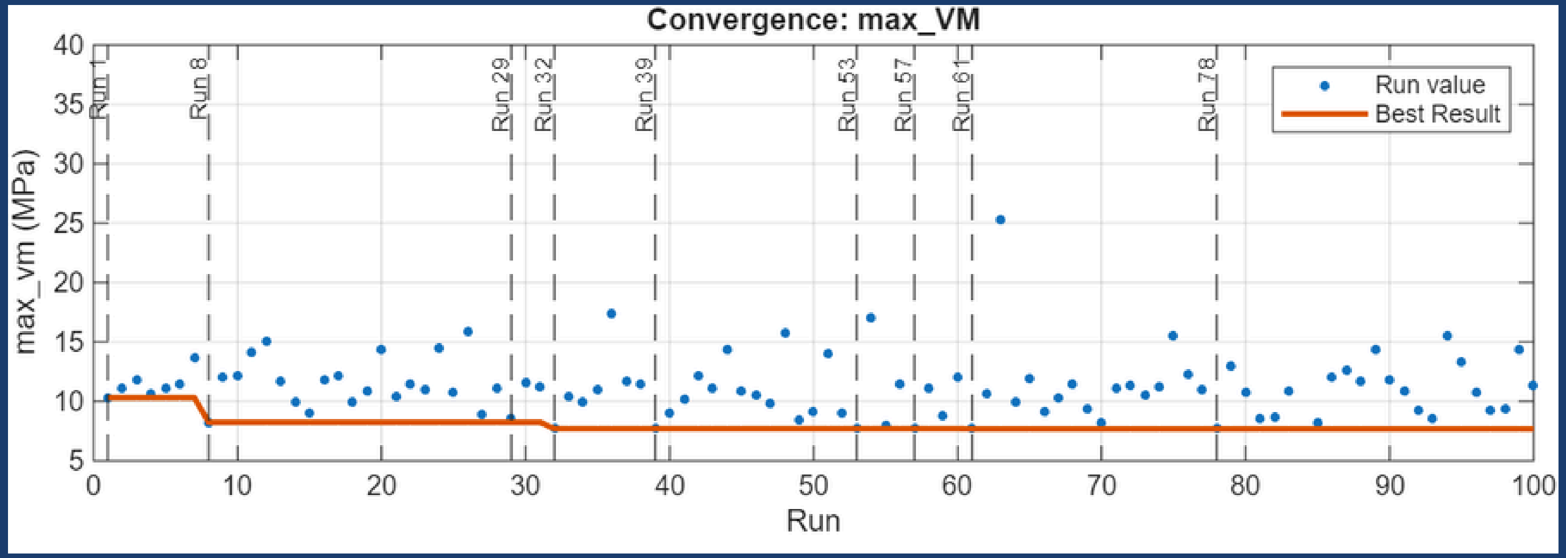


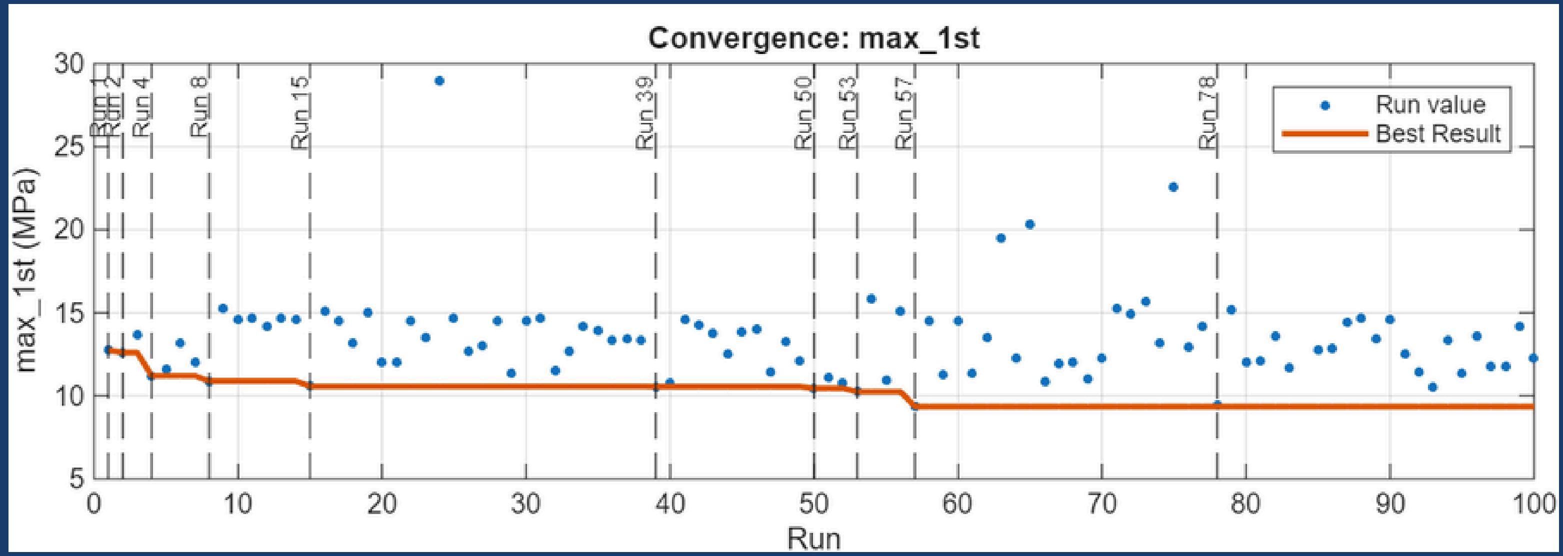


Simulation & Analysis - *Machine-learning Convergence*

- MATLAB - Statistics & Machine Learning Toolbox
  - Train on performances of configurations of unit cubes
  - Converges towards optimal combination

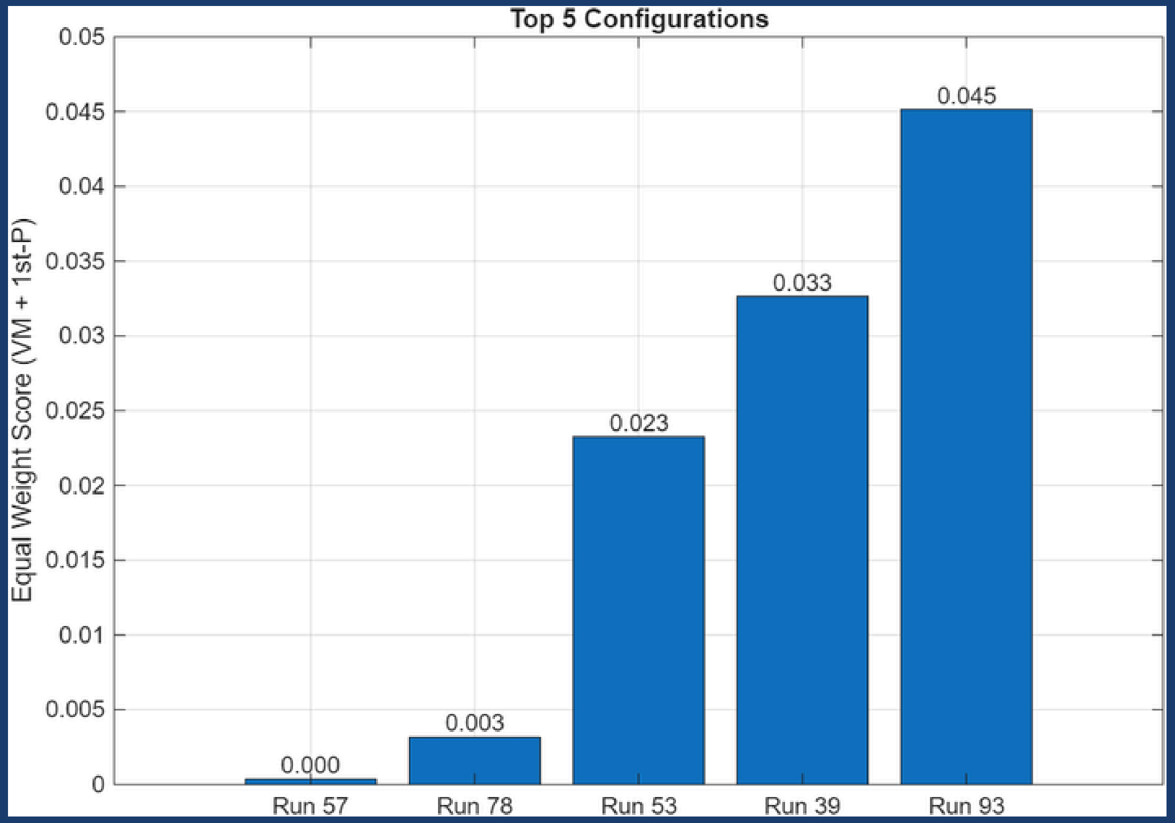
- Optimized Performance
  - VM - reduced by 26%
  - 1st - reduced by 14%

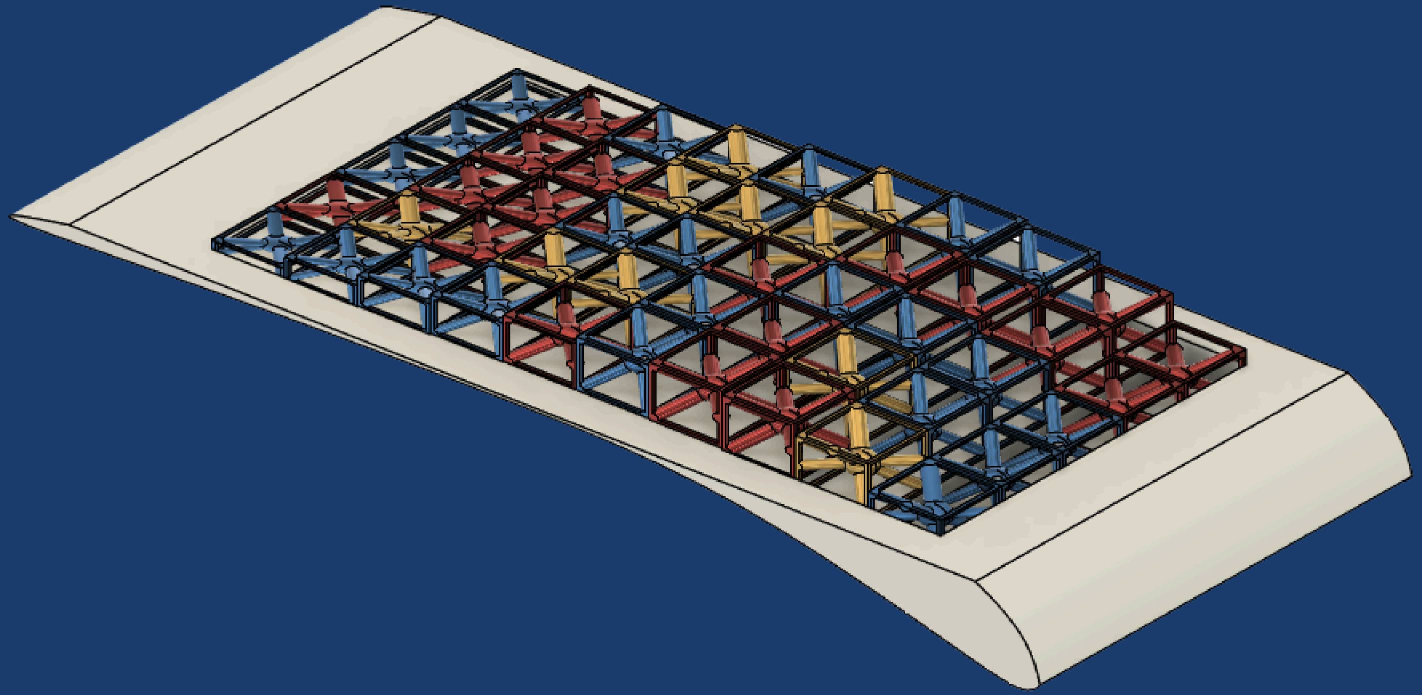


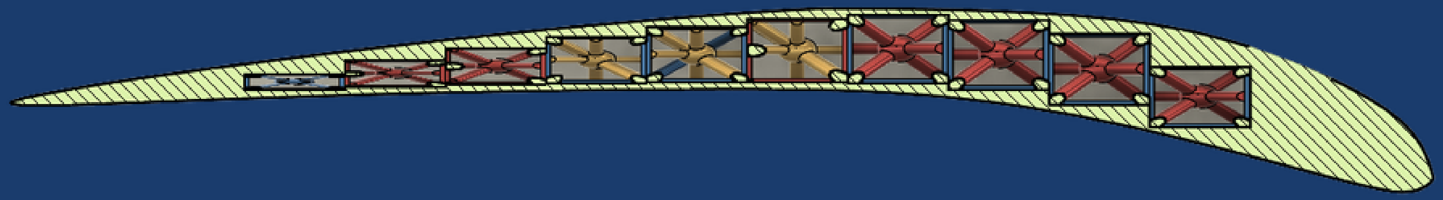


Simulation & Analysis - *Convergence Results*

- Scoring algorithm identifies best configuration between stress criteria
- Optimized result mirrors evolutionary selectrion principles
- Results in a structurally optimized lattice based on natural structures and processes







Conclusions and Recommended Next Steps

- Very low-AR wings were dominated by induced drag
- GOE 265 achieved best measured overall performance
- Stability boundaries constrained usable flight envelope
- Bio-inspired lattice improved structural survivability
- Integrated aero-structural design is the key takeaway

Next Steps

- Add repeats and uncertainty bars
- Expand angle-of-attack testing
- Compare alternate lattices experimentally

Thank You

Acknowledgment: University of Georgia wind-tunnel support